Semester 5 · Mechanical · Program Elective · 4 Credits

Power Plant Engineering

Handbook on Indian power scenario, thermal, hydro, diesel, gas turbine, combined cycle and nuclear power plants, fuels, combustion, environmental impact and power plant safety.

Course purpose

Power plant engineering explains how different power generation systems convert fuel, water energy, gas energy or nuclear energy into electricity, and how safety and environmental control must be handled.

Malayalam: Power plant □□□□□□□□□□□□ line diagram □□□□□□□ □□□□□. Fuel, boiler/turbine, generator, cooling, pollution, safety □□□□□□□ □□□□□ system □□□ □□□□□□□□□□□□□□□□.

Official details

Code: 5023C · **Title:** Power plant Engineering · **Program:** Diploma in Mechanical Engineering · **Periods:** 60.

Prerequisites: Applied Physics I and II, Thermal Engineering.

CO1

Power scenario, thermal plant, fuels and combustion.

CO2

Hydro, diesel, gas turbine and combined cycle plants.

CO3

Nuclear reaction, reactor parts and reactor types.

CO4

Environmental issues and power plant safety.

Roadmap

CO1 · 14 Hours

Power Scenario, Thermal Power Plant, Fuels and Combustion

Power plant basics

A power plant converts primary energy into electrical energy. Selection depends on fuel availability, water, load centre, transport, land, environmental rules, cooling and grid connection.

Thermal plant

Thermal power plants include coal handling, boiler, superheater, steam turbine, condenser, cooling tower, feed pump, economiser, air preheater, ash handling and generator. Reheating, bleeding and regeneration improve efficiency.

Fuels and calorific value

Good fuel should have high calorific value, low ash, safe handling, easy ignition, low cost and availability. Important properties include flash point, fire point, pour point, octane number, cetane number, HCV and LCV. Bomb calorimeter and Junkers gas calorimeter measure calorific value.

Expected questions

1. Classify power plants and explain selection factors.
2. Draw thermal power plant line diagram.
3. Explain reheating, bleeding and regeneration.
4. Explain HCV, LCV, bomb calorimeter and Junkers calorimeter.

CO2 · 15 Hours

Hydro, Diesel, Gas Turbine and Combined Cycle Plants

Hydroelectric plant

Hydroelectric plants convert water potential energy to turbine power. Layout includes dam, reservoir, intake, penstock, surge tank, turbine, draft tube, generator and tail race. Types include run-off river, storage river and pumped storage plants.

Diesel plant

Diesel power plants are useful for standby, small capacity and peak load. Layout includes fuel supply, diesel engine, air intake, exhaust, cooling, lubrication, starting system and alternator.

Gas turbine and combined cycle

Gas turbine plant includes compressor, combustion chamber, turbine and generator. Combined cycle uses gas turbine exhaust heat in a heat recovery steam generator to run a steam turbine.

Expected questions

1. Draw hydroelectric plant layout and explain.
2. Classify hydroelectric plants.
3. Draw diesel power plant layout.
4. Explain gas turbine and combined cycle plants.

CO3 · 15 Hours

Nuclear Power Plant

Nuclear reaction

Nuclear power is based on energy released during fission. Chain reaction is controlled inside a reactor. Fusion joins light nuclei but is not the common commercial plant route in this syllabus.

Reactor parts

Main parts include fuel, moderator, control rods, coolant, reflector, pressure vessel, shielding, heat exchanger/steam generator and turbine-generator system. Fuel materials include uranium, thorium and plutonium. Moderators include graphite and heavy water; coolants include water, liquid metal, gas and organic liquids.

Reactor types

BWR boils water directly in reactor. PWR keeps primary water under pressure and transfers heat through steam generator. FBR uses fast neutrons and can breed more fissile material.

Expected questions

1. Explain fission, fusion and chain reaction.
2. Draw nuclear reactor schematic.

3. Explain fuel, moderator, control rod and coolant.
4. Compare BWR, PWR and FBR.

CO4 · 14 Hours

Environmental Impact and Power Plant Safety

Environmental impact

Power plants affect air, water and land. Major issues are greenhouse gases, acid rain, acid snow, dry deposition, acid fog, fly ash, thermal pollution and radiation risk in nuclear plant effluents.

Safety policy

Plant safety needs operating procedure, permit system, isolation, PPE, fire protection, chemical handling rules, oil handling safety, boiler safety and emergency response.

Boiler and statutory provisions

Boiler operation needs pressure safety, water level control, blowdown, feedwater treatment, burner safety, safety valve and statutory inspection/compliance.

Expected questions

1. Explain greenhouse effect and acid precipitation.
2. Explain air, water and thermal pollution from power plants.
3. Explain safety practices in boiler operation.
4. Explain safety in oil and chemical handling systems.

Rule bank

Plant selection depends on fuel, water, load, grid, land and environmental clearance. Thermal efficiency improves with reheating, regeneration and waste heat recovery. Safety must be engineered, documented and inspected.

Model paper

1. Draw thermal power plant and explain components.
2. Explain hydro, diesel and gas turbine plants.
3. Explain nuclear reactor parts and reactor types.
4. Explain pollution and safety issues in power plants.

Answer guide

For plant diagrams show energy flow and label all major components. For fuels write property, meaning and importance. For safety answers include hazard, control, procedure, statutory requirement and emergency action.

References: P.K. Nag, R.S. Khurmi and J.K. Gupta, G.D. Rai, Frederick Morse, P.C. Sharma, R.K. Rajput and power plant operation resources.